संभव BATCH - CRASH COURSE

Answer Key & Solutions: Direction Sense & Syllogism (Hard)

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Zero Error Verified | Detailed Analytical Explanations (Venn Logic & Coordinates)

Question 1

Correct Answer: Exact: $5\sqrt{37}$ m

Analytical Explanation:
Coordinate Map: Let B = (0,0). A = (-15, 0). C is 10m S of B -> (0, -10). D is 20m E of C -> (20, -10). E is 15m N of D -> (20, 5). F is 5m W of E -> (15, 5). Shortest distance AF = $\sqrt{[(15 - (-15))^2 + (5 - 0)^2]} = \sqrt{(30^2 + 5^2)} = \sqrt{925} = 5\sqrt{37}$ m. (Note: SSC PYQ option anomaly).

Question 2

Correct Answer: South

Analytical Explanation:
In the morning, the Sun is in the East, so shadows fall towards the West. Vimal's shadow falls to the right of Kamal. This means Kamal's right is West, so Kamal is facing North. Since Vimal is facing Kamal, Vimal is facing South.

Question 3

Correct Answer: North (Exact)

Analytical Explanation:
Initial direction: North-West (315°). Net turn = +90° (CW) - 180° (ACW) + 135° (CW) = +45° (CW). North-West + 45° Clockwise = North. (Note: SSC options often contain misprints; the exact analytical direction is North).

Question 4

Correct Answer: Same Position (Coincide)

Analytical Explanation:
Decoding: A is 10m North of B. B is 10m East of C. C is 10m South of D. D is 10m West of E. Mapping coordinates: Let E = (0, 10), then D = (-10, 10), C = (-10, 0), B = (0, 0), A = (0, 10). Thus, A and E are at the exact same position.

Question 5

Correct Answer: 45 km East

Analytical Explanation:
Mapping displacements: Start (0,0). +20m North (0, 20). Right turn means East: +30m (30, 20). Right turn means South: +35m (30, -15). Left turn means East: +15m (45, -15). Left turn means North: +15m (45, 0). Final position is 45 km East of the start.

Question 6

Correct Answer: $\sqrt{(1249)}$ km

Analytical Explanation:
Let Y = (0,0). X is 12 km North -> X(0, 12). Z is 15 km East -> Z(15, 0). W is 20 km South of Z -> W(15, -20). Distance XW = $\sqrt{[(15 - 0)^2 + (-20 - 12)^2]} = \sqrt{(15^2 + 32^2)} = \sqrt{(225 + 1024)} = \sqrt{1249}$ km.

Question 7

Correct Answer: South

Analytical Explanation:
In the evening, the Sun is in the West, so shadows fall towards the East. Hema's shadow is exactly to her right. Therefore, Hema's right is East, meaning Hema is facing North. Since Rekha is facing Hema, Rekha faces South.

Question 8

Correct Answer: South-East

Analytical Explanation:
The damaged compass points East when it should be North (a 90° clockwise shift). Displayed Direction = Actual Direction + 90°. If Displayed = South-West (225°), Actual = 225° - 90° = 135° (South-East).

Question 9

Correct Answer: North-East

Analytical Explanation:
Coordinate Map: Let J = (0,0). K is 10m S -> K(0,-10). L is 20m E of K -> L(20,-10). M is 15m N of L -> M(20,5). N is 10m W of M -> N(10,5). N(10,5) with respect to J(0,0) is in the North-East direction.

Question 10

Correct Answer: Cannot be determined

Analytical Explanation:
Decoding: A is East of B (+x). B is South of C (-y). C is East of D (+x). D is North of E (+y). A's precise position relative to E depends entirely on the unknown numerical distances of these segments.

Question 11

Correct Answer: North

Analytical Explanation:
Path mapping: 15 km North. 10 km West. 5 km South. 10 km East. Net horizontal displacement: -10 + 10 = 0. Net vertical displacement: +15 - 5 = +10 (North). He is directly North of his house.

Question 12

Correct Answer: 20 m

Analytical Explanation:
Path mapping: Start facing South. Walks 20m S. Right turn (West) for 10m. Right turn (North) for 20m. Right turn (East) for 30m. Net vertical = -20 + 20 = 0. Net horizontal = -10 + 30 = +20m (East). Distance is 20 m.

Question 13

Correct Answer: South-East

Analytical Explanation:
South-East (135°) becomes North (0°), a shift of 135° Anti-Clockwise. West (270°) will become 270° - 135° = 135°, which corresponds to South-East.

Question 14

Correct Answer: 18 m

Analytical Explanation:
Coordinate Map: Let Q = (0,0). P = (0, 8). R = (-12, 0). S = (-12, -15). T = (0, -15). U = (0, -10). Shortest distance between P(0, 8) and U(0, -10) lies on the same vertical axis. Distance = 8 - (-10) = 18 m.

Question 15

Correct Answer: East

Analytical Explanation:
At 12 noon, both hands point to North. The clock shows North-East, meaning the entire clock is rotated 45° Clockwise. At 1:30 PM, the hour hand is exactly halfway between 1 and 2 (45° from North, i.e., North-East). Adding the 45° rotation, it points East.

Question 16

Correct Answer: South-East

Analytical Explanation:
Decoding P % Q + R - S: P is East of Q. Q is North of R. R is West of S. Let R = (0,0). Q = (0, y). S = (x, 0). S is at (+x, 0) and Q is at (0, +y). Thus, S with respect to Q is South-East.

Question 17

Correct Answer: South

Analytical Explanation:
He walks West. Turns 90° CW -> Faces North. Turns 135° ACW -> Faces South-West. Turns South (this overrides previous facing) and walks 10m. Since his last specific instruction was to 'turn South', he is facing South.

Question 18

Correct Answer: $30\sqrt{10}$ km

Analytical Explanation:
Coordinates: Let Y = (0,0). X = (50, 0). Z = (0, 30). W = (-40, 30). Distance WX = $\sqrt{[(50 - (-40))^2 + (0 - 30)^2]}$ = $\sqrt{(90^2 + (-30)^2)}$ = $\sqrt{(8100 + 900)}$ = $\sqrt{9000}$ = $30\sqrt{10}$ km.

Question 19

Correct Answer: 20 meters East

Analytical Explanation:
Path: North 15m. Right (East) 20m. Right (South) 15m. The vertical movements cancel out (+15 and -15). She is left 20m to the East of her starting point.

Question 20

Correct Answer: 65 km

Analytical Explanation:
Bus 1 effective distance on main road: 25m straight, bypasses via a rectangle (15m R, 25m L), then returns. Total forward progress = 25 + 25 = 50 km. Bus 2 traveled 35 km. Total distance covered = 50 + 35 = 85 km. Distance between them = 150 - 85 = 65 km.

Question 21

Correct Answer: West

Analytical Explanation:
Working backwards: Final direction = North. Last turn was Left. Before Left, it was East. Before that, Right turn. Before Right, it was South. Before that, Right turn. Start direction was West. (West + Right + Right + Left = North).

Question 22

Correct Answer: West

Analytical Explanation:
Coordinate Map: Let Q = (0,0). P = (-20, 0). R = (0, 15). S = (-25, 15). T = (-25, 0). P is at (-20, 0) and T is at (-25, 0). Thus, T is directly to the West of P.

Question 23

Correct Answer: 5 km

Analytical Explanation:
Net horizontal = 12E - 12W = 0. Net vertical = 5N - 10S = 5S. The drone is exactly 5 km South of the starting point. Shortest distance is 5 km.

Question 24

Correct Answer: East

Analytical Explanation:
During sunrise (Sun in East), shadows fall West. If the shadow falls exactly to the pole's right, the pole's 'right' side is West. If you look at the pole FROM the side the shadow points (from the West looking towards the pole), you are facing East.

Question 25

Correct Answer: 5m

Analytical Explanation:
Let H = (0,0). G = (0, 15). I = (10, 15). J = (10, -5). K = (0, -5). H is at (0,0) and K is at (0, -5). The distance between H and K is 5m.

Question 26

Correct Answer: Both Conclusions I and II follow

Analytical Explanation:
Venn Logic: 'Only a few Books are Pens' -> Some Books are Pens, Some are not. All Pens are Pencils. No Pencil is Marker. I is True (Books can overlap completely with Pencils outside the Pen circle). II is True (All Pens are Pencils, and Pencils cannot touch Markers, so No Pen is a Marker).

Question 27

Correct Answer: Both I and II follow

Analytical Explanation:
Venn Logic: 'Only Students are Teachers' -> All Teachers are Students (and Teachers cannot interact with any other element). Some Students are Boys. No Boy is a Girl. I is True (Teachers can only be Students, never Girls). II is True (The Students who are Boys cannot be Girls, and Teachers cannot be Girls).

Question 28

Correct Answer: Only II follows

Analytical Explanation:
Venn Logic: 'Only a few Dogs are Rats' means Some Dogs are Rats AND Some Dogs are definitely NOT Rats. Therefore, Conclusion I (All Dogs being Rats is a possibility) is False. Conclusion II is True as Cats and Mice have no negative restriction restricting their overlap.

Question 29

Correct Answer: Neither I nor II follows

Analytical Explanation:
Venn Logic: No A is B. All B are C. Only a few C are D. I is False (B and D do not have a definite positive relation). II is False (A cannot touch B, but A can exist entirely inside C without touching B).

Question 30

Correct Answer: Neither I nor II follows

Analytical Explanation:
Venn Logic: All Windows are Doors. Only a few Doors are Walls. No Wall is a Roof. I is False (Windows can overlap with Walls, it is not definitely prohibited). II is False (Since Some Doors are Walls, and No Wall is a Roof, the Doors that are Walls can never be Roofs, so All Doors cannot be Roofs).

Question 31

Correct Answer: Only II follows

Analytical Explanation:
Venn Logic: 'Only a few Red are Green' means Some Red are NOT Green, hence All Red being Green is False (I is False). All Blue are Yellow, and Some Green are Blue (from 'only a few Green are Blue'), therefore Some Green are definitely Yellow (II is True).

Question 32

Correct Answer: Both I and II follow

Analytical Explanation:
Venn Logic: No Water is Air. All Air is Fire. Some Fire is Earth. I is True (The part of Fire that is Air can never be Water). II is True (Earth can completely overlap with Air as there is no restriction).

Question 33

Correct Answer: Only I follows

Analytical Explanation:
Venn Logic: Some Laptops are Computers. No Computer is Mobile. Therefore, the Laptops that are Computers can never be Mobiles. Conclusion I is True. Conclusion II is False (Tablets include all Mobiles. Mobiles cannot be Computers, so All Tablets can never be Computers).

Question 34

Correct Answer: Only I follows

Analytical Explanation:
Venn Logic: No C is D. All D are E. This means the portion of E that contains D can never touch C. Therefore, Some E are definitely not C (I is True). A and D have no direct definite connection, so II is False.

Question 35

Correct Answer: Only I follows

Analytical Explanation:
Venn Logic: 'Only N are O' means All O are N, and O cannot overlap with any other element (exclusive). Therefore, No M is O is definitely True (I is True). Similarly, P cannot overlap with O, making II False.

Question 36

Correct Answer: Only I follows

Analytical Explanation:
Venn Logic: All Stars are Planets. Only a few Planets are Moons. I is True (Stars can completely sit inside the Moon circle). II is False (Because Some Planets are Moons, and No Moon is a Sun, the Planets that are Moons can never be Suns).

Question 37

Correct Answer: Both I and II follow

Analytical Explanation:
Venn Logic: 'Only a few Tigers are Lions' inherently means Some Tigers are NOT Lions. Thus, I is True. II is True because Cheetahs can completely envelop Tigers without violating any given negative constraints.

Question 38

Correct Answer: Both I and II follow

Analytical Explanation:
Venn Logic: No Car is a Bus. Only a few Buses are Trucks. All Trucks are Trains. I is True (The Trains that are Buses can never be Cars). II is True ('Only a few Buses are Trucks' means Some Buses are definitely not Trucks, so All Buses can never be Trucks).

Question 39

Correct Answer: Both I and II follow

Analytical Explanation:
Venn Logic: 'Only a few Apples are Mangos' means Some Apples are NOT Mangos, so All Apples can never be Mangos (I is True). 'Only a few Mangos are Bananas' means Some Mangos are Bananas. No Banana is Grapes. Thus, the Mangos that are Bananas can never be Grapes (II is True).

Question 40

Correct Answer: Only I follows

Analytical Explanation:
Venn Logic: All Keys are Locks. No Lock is a Door. Keys are entirely inside Locks, which cannot touch Doors. Thus, No Key is a Door (I is True). Some Gates are Doors. Doors cannot touch Locks. Thus, the Gates that are Doors cannot be Locks, making Conclusion II False.

Question 41

Correct Answer: Only I follows

Analytical Explanation:
Venn Logic: Only a few Circles are Squares. All Squares are Triangles. I is True (Circles can completely overlap with Triangles, outside the Square boundary). II is False (No definite positive relationship is established between Rectangles and Squares).

Question 42

Correct Answer: Only II follows

Analytical Explanation:
Venn Logic: Only a few A are B. Only a few B are C. No C is D. I is False (A and C have no definite negative relation). II is True (The B that are C can never be D because No C is D).

Question 43

Correct Answer: Only I follows

Analytical Explanation:
Venn Logic: 'Only 4 are 5' means All 5 are 4, and 5 cannot interact with 1, 2, or 3. Thus, No 3 is 5 is definitely True (I is True). 1 and 3 do not have a definite positive overlap, so II is False.

Question 44

Correct Answer: Both I and II follow

Analytical Explanation:
Venn Logic: 'Only a few Papers are Files' means Some Papers are definitely not Files, making I True. There is no restriction preventing all Folders from being Papers, so II is a valid possibility (True).

Question 45

Correct Answer: Only II follows

Analytical Explanation:
Venn Logic: Some White are Black. Only a few Black are Red. No Red is Blue. I is False (White and Blue have no definite negative relation). II is True (The Black that are Red can never be Blue, so All Black can never be Blue).

Question 46

Correct Answer: Both I and II follow

Analytical Explanation:
Venn Logic: 'Only a few Men are Women' means Some Men are NOT Women (I is True). There is no negative restriction preventing Women from completely overlapping with Adults, so II is a valid possibility (True).

Question 47

Correct Answer: Only I follows

Analytical Explanation:
Venn Logic: No One is Two. Only a few Two are Three. All Three are Four. I is True (The Four that are Two can never be One). II is False ('Only a few Two are Three' guarantees Some Two are NOT Three, so All Two can never be Three).

Question 48

Correct Answer: Only I follows

Analytical Explanation:
Venn Logic: Only a few X are Y. All Y are Z. I is True (X can completely exist inside Z, as long as it doesn't completely enter Y). II is False (Y and W have no definite positive relation).

Question 49

Correct Answer: Both I and II follow

Analytical Explanation:
Venn Logic: All Alpha are Beta. No Beta is Gamma. I is True (Since Alpha is entirely inside Beta, and Beta cannot touch Gamma). II is True (The Delta that are Gamma can never be Beta).

Question 50

Correct Answer: **Only II follows**

Analytical Explanation:
Venn Logic: Only a few Cups are Plates. No Plate is a Spoon. Some Spoons are Forks. I is False (The Cups that are Plates can never be Spoons, so All Cups cannot be Spoons). II is True (The Forks that are Spoons can never be Plates).

End of Answer Key - "संभव Batch" - Prepared for Excellence with Zero Errors.